MIL-I-7065A
26 May 1976
Superseding
MIL-I-7065
28 December 1950

MILITARY SPECIFICATION

INDICATOR, POSITION, AILERON TRIM TAB

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers one type of aileron trim tab position indicator.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents.- The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

Specifications

Federal

PPP-B-636	Box Fiberboard

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-T-5583	Transmitter; Position, 28-Volt DC
MIL-I-7062	Indicators, Position, Control Surfaces, 28 Volt DC, General Specification for

Standards

Military

MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

FSC 6610

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:54Z
Check the source to verify that this is the current version before use.

Military

MS-28002	Indicator-Position, Control Surfaces, 1-3/8 Inch Dial, 28 Volt DC
MS-33678	Connector, Receptacle, Electric, Integral Mounting
MS-33680	Insert Arrangements, MIL-C-5015 Electric Connector, Sizes 8 and 10

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1. Qualification.- The Position Indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see para 6.2).

3.2. General.- The requirements specified in Specification MIL-I-7062 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.2.1 In the event the requirements of this specification and the general specification conflict, the requirements of this specification shall govern.

3.3 Design and Construction.- The indicator shall be designed for operation with a position transmitter, in accordance with Specification MIL-T-5583.

3.3.1 Resistance.- The direct-current resistance of the indicator measured between the connector pins A, B, and C shall be 475 ±100 ohms. Deviation from the average of the resistance values between the three coils shall not exceed ±1-1/2 percent.

3.3.2 Pointer.- The pointer shall conform to the dimensions shown on figure 1. The length of the pointer shall be such that the tips of the pointer shall extend into the scale from 1/3 to 2/3 the length of the shortest graduation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:54Z
Check the source to verify that this is the current version before use.

3.3.3 Electrical connector.- The indicator shall be provided with a MS33678-10SL-3P electrical connector in accordance with MIL-I-7062 for hermatically sealed indicators.

3.3.4 Indicator Case.- The case shall conform to Standard MS28002. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The design of the case shall be such that the internal mechanism may be removed, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures unless they are approved by the Procuring Service. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected.

3.3.5 Dial.- The dial shall conform to figure 1. Each graduation shall be equivalent to a 5 indicated degree interval. The dial scale plate shall be securely fastened to the frame of the mechanism. The minimum diameter of the dial shall be 1-5/16 inches when measured across the outside ends of the graduations.

3.4 Markings.- The dial markings shall be as shown on figure 1, and shall be durable to withstand usage encountered in service.

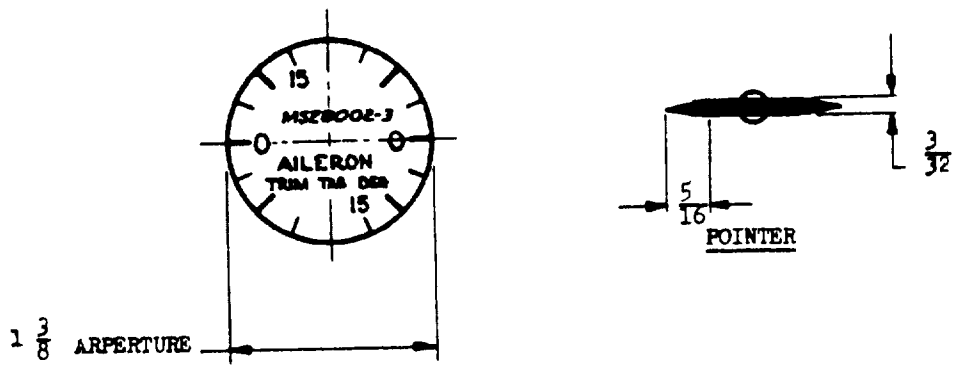
3.5 Identification of Product.-

3.5.1 Nameplate.- A nameplate, permanently and legibly filled in with the following information, shall be securely attached to the indicator:

Indicator, Position, Aileron Trim Tab, 28-Volts DC
MS Part No.
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract of Order No.
Manufacturer's Name or Trademark
U S Property

3.5.2 Use of AN or MIL Designations.- An or MIL designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence or sales matter, until notification has been received from the Qualifying Service that the product has been approved for aeronautical use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:54Z
Check the source to verify that this is the current version before use.



MARKING	HEIGHT OR LENGTH ±0.010	WIDTH OF LINE OR GRADUATION ±0.005	MATERIAL OR FINISH
LETTERING "AILERON" NUMERAL "0" NUMERAL "15" LETTERING "TRIM TAB DEG" MAJOR GRADUATIONS MINOR GRADUATIONS SHADED PROTION OF POINTER	0.094 0.125 0.094 0.062 0.156 0.125 - ---	0.020 0.020 0.020 0.020 0.031 0.020 - ---	FLUORESCENT LUMINESCENT
LETTERING "MS28002 -3" BACKGROUND OF DIAL	0.062 - ---	0.010 - ---	DURABLE DULL BLACK

DIMENSIONS IN INCHES

FIGURE 1. Dial, Pointer, and Markings

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:54Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality Assurance Provisions shall be as specified in Specification MIL-I-7062, except as follows:

4.1.1 Voltage Off.- With no voltage applied to the indicator system, the pointer shall move off scale to an approximate vertical position.

4.2 Inspection and Test.- Tests for Methods of Preservation and Packaging will be accomplished in accordance with Section 4 of MIL-P-116 to assure compliance with Section 5 of this specification.

5. Packaging.

5.1 Preservation-Packaging.- Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning.- Indicators shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying.- Indicators shall be dried in accordance with Process D-4 of MIL-P-116.

5.1.1.3 Preservation Application.- Not applicable.

5.1.1.4 Unit Packaging.- Unless otherwise specified by the procuring activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method 1C1 of MIL-P-116. Overbox completed packs in PPP-B-636, weather-resistant class. Cushioning material between bag and unit container to be of a type, density and thickness that will insure shock transmission does not exceed the peak values in G's established for the item, when completed packs are subjected to the rough handling drop tests stipulated by MIL-STD-794.

5.1.2 Level C.- Indicators shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packaging.- Packing shall be level A, B, or C as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:54Z
Check the source to verify that this is the current version before use.

5.2.1 Level A.- Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, weather-resistant class. Insofar as possible, shipping containers will be uniform shape, size, minimum tare and cube consistant with protection required.

5.2.2 Level B.- Indicators packaged as specified in paragraph 5.1.1 shall be packed in PPP-B-636 Box, fiberboard, domestic class. For the Army, the exterior container shall be selected from Table II, MIL-STD-794, for the applicable level of packing.

5.2.3 Level C.- Packing shall be applied which affords adequate protection against physical and mechanical damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's standard commercial practice.

5.3 Marking.

5.3.1 Container Markings.- Unit packages and shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows: Indicator, Position, Aileron Trim Tab.

6 NOTES

6.1 Intended Use.- The aileron trim tab position indicator covered by this specification is intended for use with a position transmitter, conforming to Specification MIL-T-5583, in indicating the position of the aircraft component in degree travel from the neutral position, and the reaction on the wings of the aircraft such as tending to lower the right wing or the left wing.

6.1.1 For trim tabs, the AN 5786 position transmitter is connected to the tab actuating mechanism so that a 2 to 1 movement ratio is provided. The transmitter shaft shall be rotated 2 mechanical degrees for each degree of tab travel. Example: If a tab travels 15 degrees from the neutral or zero position to 1 extreme deflection, the transmitter shall be connected so that the transmitter shaft rotates 30 degrees for the 15 degrees tab travel.

6.1.2 The trim tab indicator may be used with the AN 5787 position transmitter which has the same electrical characteristics as in the AN 5786 position transmitter. The transmitter brushes must be rotated 4 mechanical degrees for each 1 degree of tab travel

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:54Z
Check the source to verify that this is the current version before use.

6.2 Qualification.- With respect to products requiring qualifications, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City ALC, MMEO, Tinker AFB, Oklahoma, and information pertaining to qualification of products may be obtained from that activity.

6.2.1 It is to be understood that upon receipt of the Letter of Authorization, samples shall be furnished at no cost to the Government, and that the manufacturer shall pay the transportation charges to and from the designated point where tests are to be made. In the case of failure of the sample or samples submitted, consideration will be given to the request of the manufacturer for additional tests only after it has been clearly shown that changes have been made in the product which the Government considers sufficient to warrant additional tests.

Custodian:
Army - AV
Navy - AS

User:
Army - MI

Preparing Activity
Air Force - 71

Project No:
6610-0391

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

OMB Approval No. 22-R295

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 **1426**

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

C 1600.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:54Z
Check the source to verify that this is the current version before use.

FOLD

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE
DoD-318

U.S. MAIL

Oklahoma City ALC/MMEO
Tinker AFB OK 73145

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:54Z
Check the source to verify that this is the current version before use.